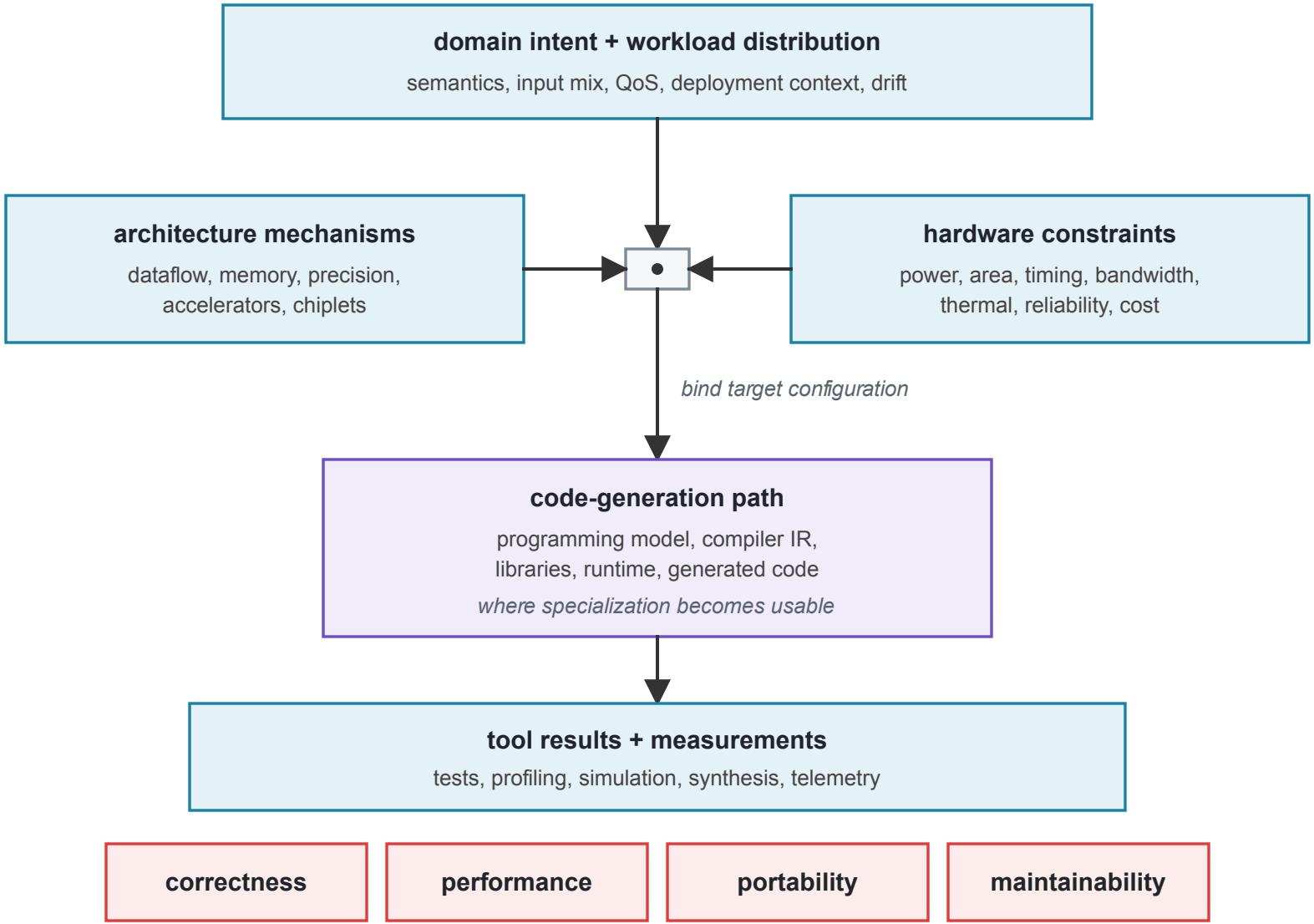
domain intent + workload distribution
semantics, input mix, QoS, deployment context, drift
architecture mechanisms
dataflow, memory, precision,
accelerators, chiplets
hardware constraints
power, area, timing, bandwidth,
thermal, reliability, cost
bind target configuration
code-generation path
programming model, compiler IR,
libraries, runtime, generated code
where specialization becomes usable
tool results + measurements
tests, profiling, simulation, synthesis, telemetry
correctness
performance
portability
maintainability